

Tutorial Book



B.Sc Drive Technology

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This TutorialBook is a
complement to it's corresponding Lecture.
When in doubt regarding theory, please consult the LectureBook.



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1

Magnetic Circuits and Materials

Q1

A Simple Magnetic Circuit

The magnetic circuit shown below has dimensions $A_c = 9 \times 10^{-4} \text{ m}^2$, $A_g = 9 \times 10^{-4} \text{ m}^2$, $g = 5 \times 10^{-4} \text{ m}$, $l_c = 0,3 \text{ m}$, and $N = 500$.

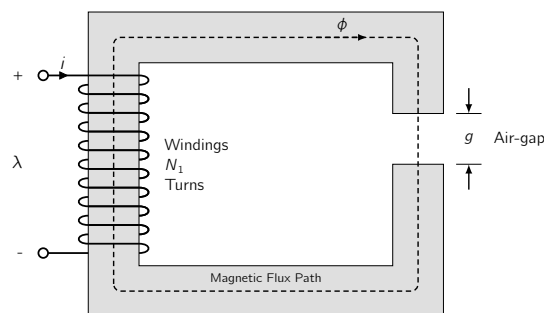


Figure 1.1

Assume the value $\mu_r = 70000$ for core material:

- Find the reluctances R_c and R_g .

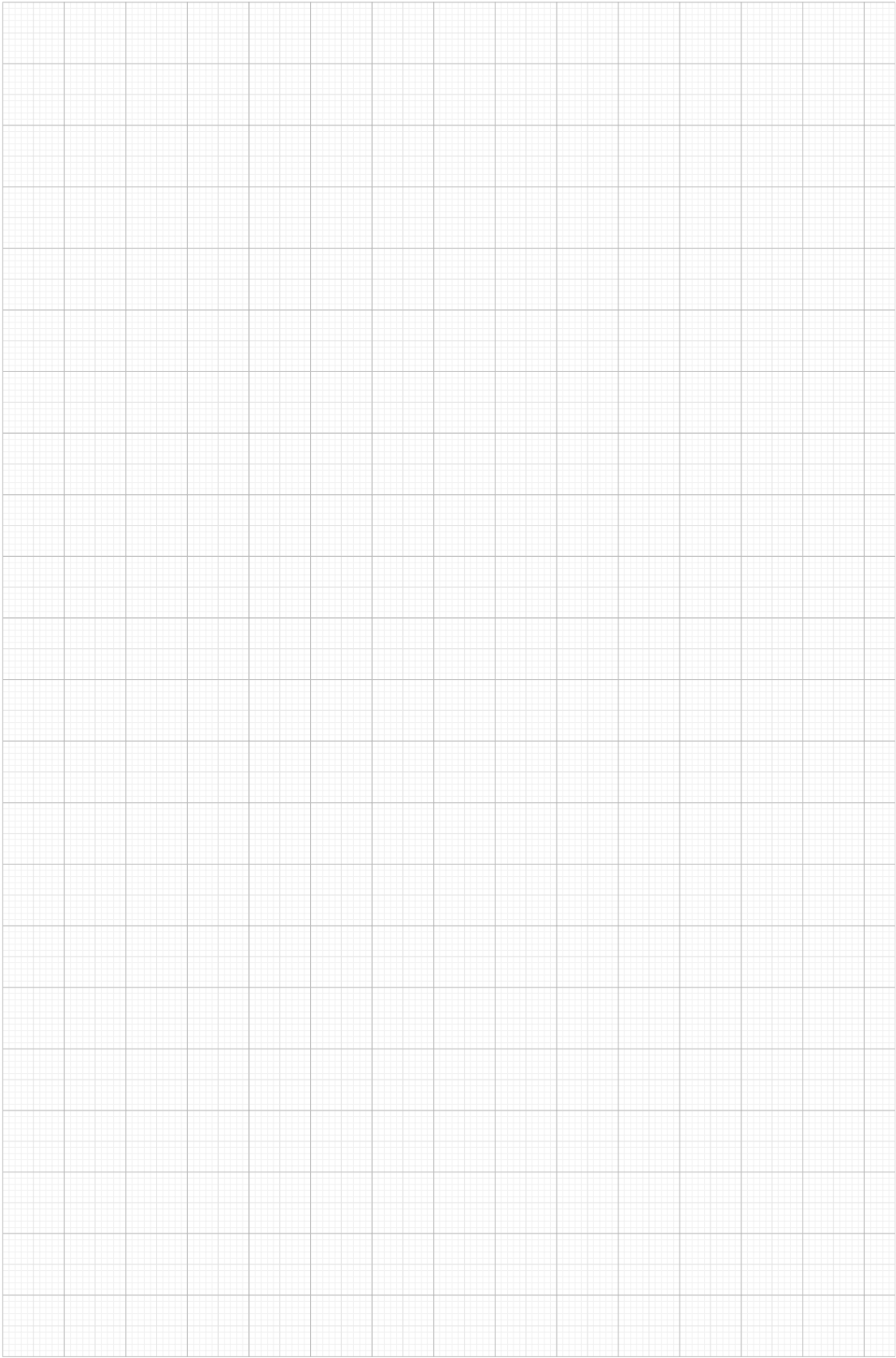
For the condition that the magnetic circuit is operating with $B_c = 1 \text{ T}$,

- Find the flux (ϕ)
- Find the current (i)

(a) $3,789,403 \times 10^3 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$, $4,420,970,64 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$,
(b) $9 \times 10^{-4} \text{ Wb}$, (c) $0,803 \text{ A}$.

Answer





Q2

A Simple Machine

The magnetic structure of a synchronous machine is shown schematically below. Assuming both rotor and stator have infinite permeability ($\mu \rightarrow \infty$), find the air-gap flux (ϕ) and flux density B_g . For this example $I = 10$ A, $N = 1,000 \times 10^3$ Turn, $g = 0,01$ m, and $A_g = 0,02$ m².

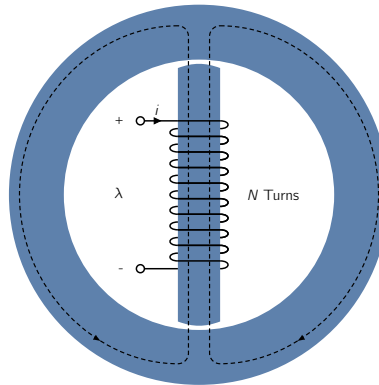
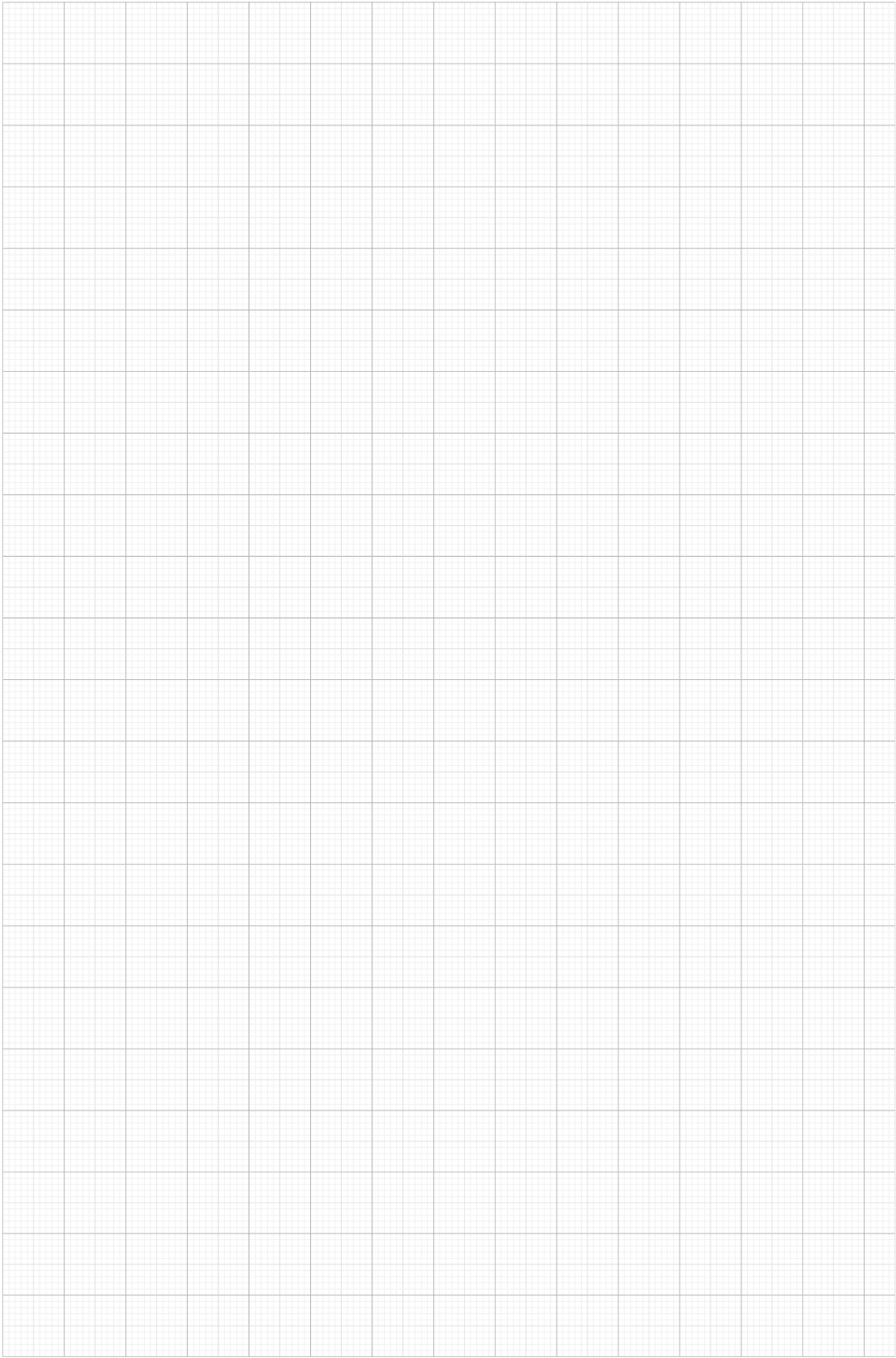


Figure 1.2: A simple synchronous machine, the air gap has a length of g and both the stator and rotor have infinite permeability.

$$B_g = 0.6285 \text{ T}$$

Answer





Q3

Analysing a Magnetic Circuit

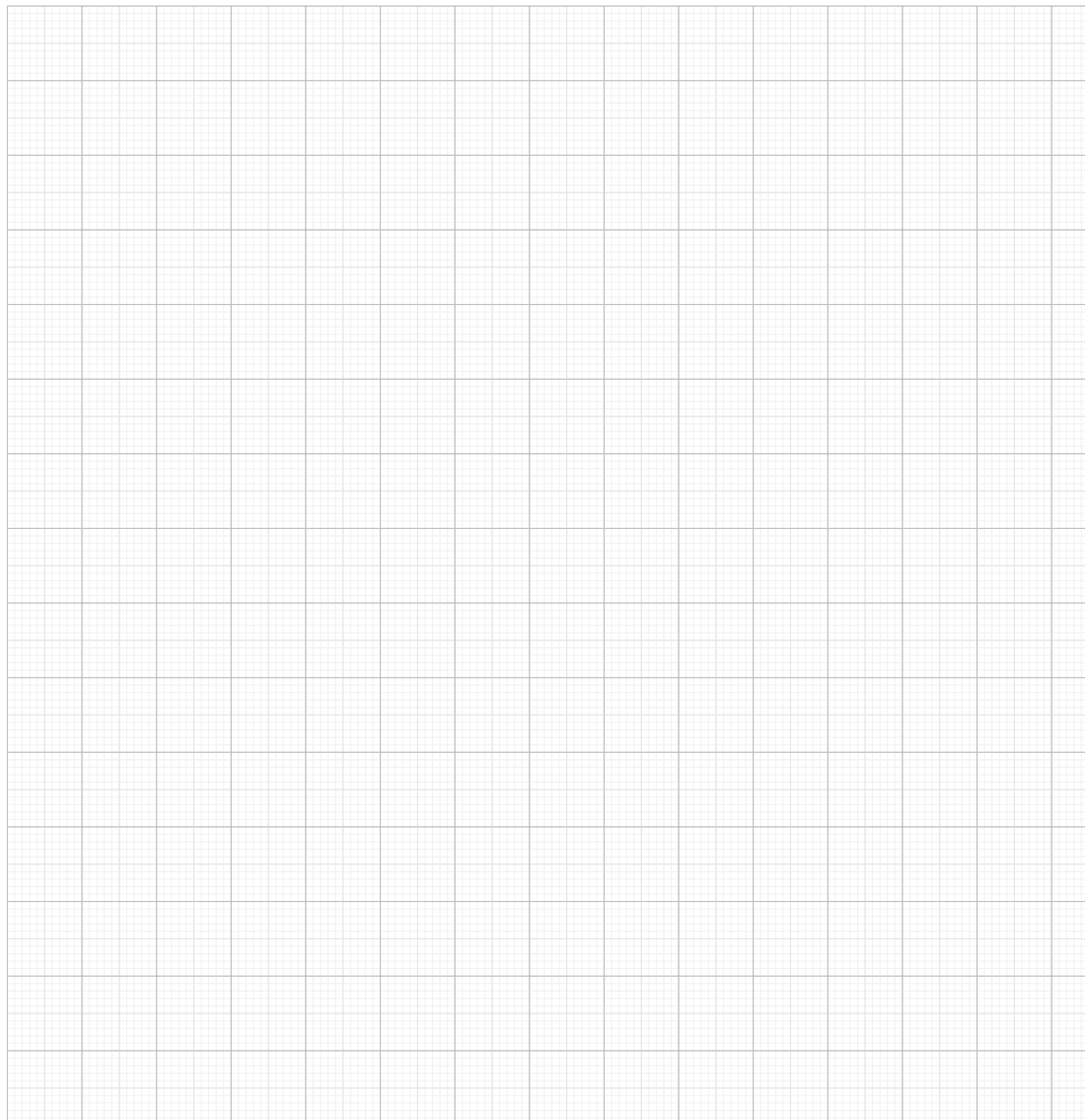
The magnetic circuit of Fig. 1.6a consists of an N -turn winding on a magnetic core of infinite permeability with two parallel air gaps of lengths g_1 and g_2 and areas A_1 and A_2 , respectively.

- the inductance of the winding,
- the flux density B_1 in gap 1 when the winding is carrying a current i .

Note Neglect fringing effects at the air gap.

$$\frac{1}{\mu_0 N^2} = \frac{1}{\phi} = \frac{1}{B_1} = \frac{1}{\mu_0} \left(\frac{g_1}{A_1} + \frac{g_2}{A_2} \right) \quad (\mathbf{a}) \cdot L = \mu_0 N^2 \left(\frac{g_1}{A_1} + \frac{g_2}{A_2} \right) \quad (\mathbf{b})$$

Answer



Q4

Magnetic Circuit with Core Permeability

In **Ex. 1**, the relative permeability of the core material for the magnetic circuit below is assumed to be $\mu_r = 7,0000 \times 10^4$ at a flux density of 1 T.

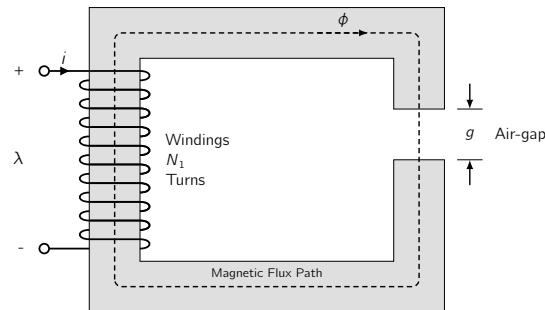


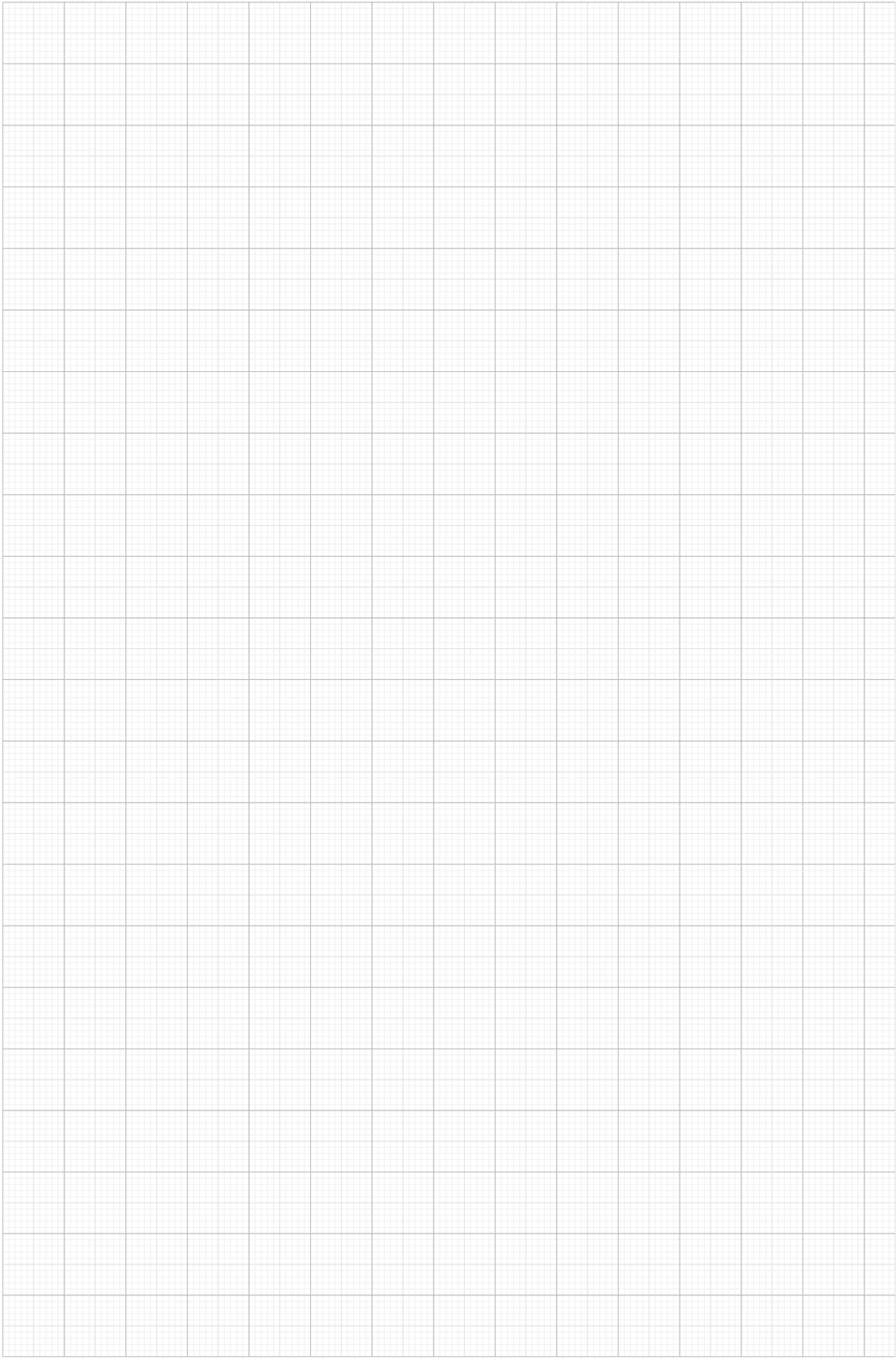
Figure 1.3: The magnetic circuit in question.

- In a practical device, the core would be constructed from electrical steel such as M-5 electrical steel. This material is highly non-linear and its relative permeability varies from a value of approximately $\mu_r = 7,2300 \times 10^4$ at a flux density of $B = 1$ T to a value of on the order of $\mu_r = 2,900 \times 10^3$ as the flux density is raised to 1,8 T. Based on this information, please calculate the inductance under the assumption that the relative permeability of the core steel is $7,2300 \times 10^4$.
- Calculate the inductance under the assumption that the relative permeability (μ_r) is equal to $2,900 \times 10^3$.

(a) 0.561 H, (b) 0.468 H

Answer





Q5

Plotting Permeability

Using Python, plot the inductance of the magnetic circuit given in **Ex. 1** as a function of core permeability over the range $100 \leq \mu_r \leq 1,000.00 \times 10^5$.



Q6

Time Varying Voltage in Magnetic Circuit

For the magnetic circuit of **Ex. 1** with the following diagram:

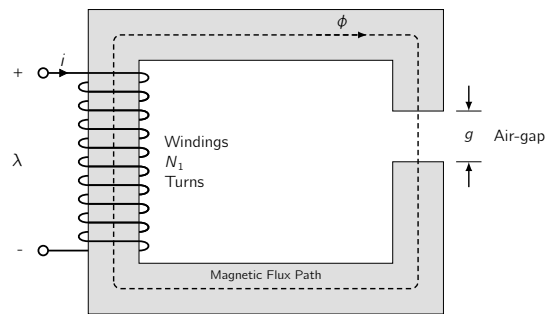


Figure 1.4: The circuit in question.

Find:

1. the inductance (L),
2. the magnetic stored energy W for $B_c = 1 \text{ T}$.
3. the induced voltage e for a 60 Hz time-varying core flux of the form $B_c = 1 \sin \omega t$.

$$(a) 0.56068 \text{ H}, (b) 0.18077 \text{ J}, (c) (377 \times 1.0 \cos(377t))$$

Answer

